Q1)

```
public class Q1 {

    static <T extends Number> double sum(T[] arr) {
        double total = 0;

        for (T x : arr) {
            total += x.doubleValue();
        }

        return total;
    }

    public static void main(String[] args) {

        Integer[] a = {1, 2, 3, 4};
        Float[] b = {1.5f, 2.5f, 3.5f};
        Double[] c = {1.2, 2.3, 3.4};

        System.out.println(sum(a));
        System.out.println(sum(b));
        System.out.println(sum(c));
    }
}
```

Q2)

```
import java.util.*;

public class Q2 {
```

```
    public static void main(String[] args) {

        Optional<String> email = Optional.of("student@gmail.com");

        Optional<String> empty = Optional.empty();

        email.ifPresent(x -> System.out.println("Email: " + x));

        System.out.println(empty.orElse("default@gmail.com"));

        try {
            System.out.println(empty.orElseThrow());
        } catch (Exception e) {
            System.out.println("Email is missing");
        }
    }
}
```

Q3)

```
import java.util.*;

class Result<T extends Number> {

    T marks;

    Result(T marks) {
        this.marks = Objects.requireNonNull(marks);
    }

    double percentage() {
```

```
        return marks.doubleValue();
    }

    boolean passed(double threshold) {
        return percentage() >= threshold;
    }

    void display() {
        Optional<T> result = Optional.ofNullable(marks);

        result.ifPresent(x -> {
            System.out.println("Marks: " + x);
            System.out.println("Percentage: " + percentage());
        });
    }
}

public class Q3 {

    public static void main(String[] args) {

        Result<Integer> r = new Result<>(75);

        r.display();

        System.out.println("Passed: " + r.passed(40));
    }
}
```

Q4)

```
import java.time.*;
import java.util.*;

public class Q4 {

    public static void main(String[] args) {

        String eventName = "College Fest";
        LocalDate eventDate = LocalDate.now().plusDays(5);
        LocalTime startTime = LocalTime.of(10, 0);

        Optional<String> venue = Optional.of("Main Auditorium");

        System.out.println("Event: " + eventName);
        System.out.println("Date: " + eventDate);
        System.out.println("Start Time: " + startTime);

        if (eventDate.isAfter(LocalDate.now()))
            System.out.println("Event is in the future");
        else if (eventDate.equals(LocalDate.now()))
            System.out.println("Event is today");
        else
            System.out.println("Event is in the past");

        System.out.println("Days remaining: " +
                (eventDate.toEpochDay() - LocalDate.now().toEpochDay()));

        System.out.println("End Time: " + startTime.plusHours(2));

        System.out.println("Venue: " +
                venue.orElse("Venue not specified"));
```

```
    }
}
```

Q5)

```
import java.util.*;

class Utility {

    static <T> boolean isEqual(T a, T b) {
        return Objects.equals(a, b);
    }

    static <T> T getDefault(Optional<T> value, T defaultValue) {
        return value.orElse(defaultValue);
    }

    static <T extends Comparable<T>> T max(T a, T b) {
        return a.compareTo(b) > 0 ? a : b;
    }

    static boolean isNull(Object obj) {
        return Objects.isNull(obj);
    }
}

public class Q5 {

    public static void main(String[] args) {
```

```
        System.out.println(Utility.isEqual("Hello", "Hello"));

        Optional<String> name = Optional.empty();
        System.out.println(
            Utility.getDefault(name, "Default Name")
        );

        System.out.println(Utility.max(10, 20));

        System.out.println(Utility.max("Apple", "Banana"));

        System.out.println(Utility.isNull(null));
    }
}
```